

Homework 1

● Graded

Student

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Total Points

49 / 50 pts

Question 1

Question 1

10 / 10 pts

1.1 Part (a)

5 / 5 pts

✓ + 5 pts Correct

+ 4 pts Calculation error, right procedure.

+ 3 pts Right procedure, but there are some unjustified steps.

+ 2 pts Right procedure, but makes numerous leaps of logic

+ 1 pt Sets up the problem. Procedure is fundamentally wrong.

+ 0 pts Missing or Incorrect without procedure.

1.2 Part (b)

5 / 5 pts

✓ + 5 pts Correct

+ 4 pts Calculation error, right procedure.

+ 3 pts Right procedure, but some unjustified steps.

+ 2 pts Right Procedure, but numerous leaps in logic.

+ 1 pt Sets up the problem. Fundamentally wrong procedure.

+ 0 pts Missing or Incorrect without procedure.

Question 2

Question 2

10 / 10 pts

2.1 Part (a)

2 / 2 pts

✓ + 2 pts Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

2.2 Part (b)

2 / 2 pts

✓ + 2 pts Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

2.3 Part (c)

3 / 3 pts

✓ + 3 pts Correct

+ 2 pts Minor mistakes, but the logic is mostly sound.

+ 1 pt Only sets up the problem.

+ 0 pts Missing or essentially incorrect.

2.4 Part (d)

3 / 3 pts

✓ + 3 pts Correct

+ 2 pts Minor mistakes, but the logic is mostly sound.

+ 1 pt Only sets up the problem.

+ 0 pts Missing or essentially incorrect.

Question 3

Question 3

10 / 10 pts

✓ + 10 pts Correct

+ 8 pts Minor mistakes, but the logic is mostly sound.

+ 6 pts Justifies most steps, but makes numerous logical leaps.

+ 4 pts Arrives at an answer, with little to no justification.

+ 2 pts Only the set up of the problem has been written.

+ 0 pts Missing or essentially incorrect.

3

avoid the cd notation.

Question 4

Question 4

9 / 10 pts

4.1 Part (a)

5 / 5 pts

✓ + 5 pts Correct

- + 4 pts Minor errors, mostly sound.
- + 3 pts Minor or no errors, but very messy solution.
- + 2 pts Procedure is essentially incorrect.
- + 1 pt Sets up the problem. No complete solution.
- + 0 pts Missing or essentially incorrect.

4.2 Part (b)

4 / 5 pts

+ 5 pts Correct

✓ + 4 pts Minor errors, mostly sound.

- + 3 pts Numerous, important leaps of logic.
- + 2 pts Essentially incorrect procedure.
- + 1 pt Sets up the problem. No complete solution.
- + 0 pts Missing.

1 You forgot the powers on the primes.

2 Again, if you had written out the powers, you would have gotten that $q = 1$, and be done with it all.

Question 5

Question 5

10 / 10 pts

5.1 — **Part (a)**

2 / 2 pts

✓ **+ 2 pts** Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

5.2 — **Part (b)**

2 / 2 pts

✓ **+ 2 pts** Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

5.3 — **Part (c)**

2 / 2 pts

✓ **+ 2 pts** Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

5.4 — **Part (d)**

2 / 2 pts

✓ **+ 2 pts** Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

5.5 — **Part (e)**

2 / 2 pts

✓ **+ 2 pts** Correct

+ 1 pt Minor logical mistakes.

+ 0 pts Missing or essentially incorrect.

Question assigned to the following page: [1.1](#)

1. A **square** (respectively **cubic**) integer is an integer that factors as a^2 (respectively a^3) for $a \in \mathbb{Z}$.

1. Prove that when a square integer is divided by 3, the remainder can never be 2.

2. Prove that a cubic integer can be written in the form $9k$, $9k+1$, or $9k-1$ for some $k \in \mathbb{Z}$.

Pf. 1. We prove by cases.

By Corollary 2.5, $\mathbb{Z}$ is separated by $[1]_3$, $[2]_3$ and $[0]_3$ exactly,

So any integer must be either $3k$, $3k+1$ or $3k+2$ for some integer k .

Let a be an arbitrary integer

So a^2 is an arbitrary square integer.

Case 1 $a = 3k$ for some $k \in \mathbb{Z}$

Then $a^2 = 9k^2 = 3(3k^2)$, $3|a^2$.

Case 2 $a = 3k+1$ for some $k \in \mathbb{Z}$

Then $a^2 = (3k+1)^2 = 3(3k^2+2k)+1$
the remainder when a^2 divided
by 3 is 1.

Case 3 $a = 3k+2$ for some $k \in \mathbb{Z}$.

Then $a^2 = (3k+2)^2 = 3(3k^2+4k)+1$
the remainder when a^2 divided
by 3 is 1.

Questions assigned to the following page: [1.1](#) and [1.2](#)

Since the 3 cases cover all situations and in none of them the remainder is 2, we have completed the proof.

2. Let a be an arbitrary integer.

Then a^3 is an arbitrary cubic integer.

Same as part I, we divide all situations into 3 cases:

Case 1 $a = 3m$ for some integer m .

$$\text{Then } a^3 = 27m^3 = 9(3m^3)$$

$$\text{So take } k = 3m^3, a^3 = 9k$$

Case 2 $a = 3m + 1$ for some integer m

$$\text{Then } a^3 = 27m^3 + 27m^2 + 9m + 1$$

$$\text{So take } k = 3m^3 + 3m^2 + m, a^3 = 9k + 1$$

Case 3 $a = 3m + 2$ for some integer m .

$$\text{Then } a^3 = 27m^3 + 54m^2 + 36m + 8$$

$$\text{So take } k = 3m^3 + 6m^2 + 4m + 1, \\ a^3 = 9k - 1$$

Since this covers all situations, we have proved the statement.

Questions assigned to the following page: [2.1](#) and [2.2](#)

2. Analogous to the idea of "greatest common divisor" is the idea of "least common multiple." The least common multiple of two (positive) integers, a and b , is the smallest (positive) integer m such that $a|m$ and $b|m$.
- Use the well-ordering principle to prove that the least common multiple of two integers always exists.
 - Let $d = \gcd(a, b)$. Prove that there is some integer m such that $dm = ab$.
 - Prove that m from part (b) is divisible by a and divisible by b .
 - Prove that for all integers M such that M is a common multiple of a and b , the number $m = \frac{ab}{\gcd(a, b)}$ (from parts b and c) divides M , making m the least common multiple. (Hint: prove that $\frac{M}{m}$ is an integer using Bezout's identity).

(a) Pf Let a, b be arbitrary positive integers.
 Let $S = \{s \in \mathbb{Z}^+ \mid a|s \text{ and } b|s\}$

First, $ab \in S$, so S is non-empty

Also, since any element $s \in S$ is also an element of $\mathbb{Z}^+$, $S \subseteq \mathbb{Z}^+ \subseteq \mathbb{Z}_{\geq 0}$

So by well-ordering principle, S has a smallest element, then by definition, the least common multiple of a, b exists.

Since a, b is arbitrary, we have proved the statement.

(b) Pf Since $d|a, d|b$ by definition of $\gcd$
 $a = dp, b = dq$ for some $p, q \in \mathbb{Z}$

So $ab = dpq$. Take pq as m , then $dm = ab$.

Questions assigned to the following page: [2.3](#) and [2.4](#)

(c) Since $dm = ab$, $a = dp$

we have $dm = dpb$

$$m = pb \text{ where } p \in \mathbb{Z} \text{ in } (b)$$

Similarly, we have $m = qa$ where $q \in \mathbb{Z} \text{ in } (b)$

So $a|m$, $b|m$.

(d) Let M be an arbitrary common multiple of a, b . Then $M = sa = kb$ for some integers s, k .

By Bezout, $(a, b) = ra + sb$ for some integers r, s .

$$\text{So } m = \frac{ab}{(a, b)} = \frac{ab}{ra + sb}, \quad \frac{M}{m} = \frac{ra + sb}{\frac{ab}{m}} \cdot kb \\ = rk + \frac{ksb}{a}$$

$$\text{Since } kb = sa, \quad \frac{M}{m} = rk + \frac{s^2a}{a} = rk + s^2$$

Since $r, k, s \in \mathbb{Z}$, $\frac{M}{m}$ is an integer
so $m|M$.

And this implies for all multiple of a, b

$$M, \quad m \leq M. \text{ Also, } m = \frac{ab}{(a, b)} = k_1 b = k_2 a$$

for some integers k_1, k_2 since $(a, b) | a$,

$(a, b) | b$. Therefore m is a common multiple of a, b , and is the least common multiple.

Question assigned to the following page: [3](#)

3. Let $a, b, c, d \in \mathbb{Z}$ such that the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

has an inverse whose entries are all integers. Let $x, y \in \mathbb{Z}$ such that at least one of them is not zero. Prove that $\gcd(x, y) = \gcd(ax + by, cx + dy)$.¹

¹How can we relate x and y with $ax + by$ and $cx + dy$ using the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$?

Denote $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ by A and its inverse by A^{-1} .

Note that $\det A = ad - bc \in \mathbb{Z}$, and $\det A^{-1}$

Therefore $\det A = ad - bc = 1$ or -1 , $\det A^{-1} = \frac{1}{\det A} \in \mathbb{Z}$.

$$\text{So } A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (\pm 1) \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\text{Then } A^{-1} \cdot \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} = \begin{pmatrix} (ad-bc)x \\ (ad-bc)y \end{pmatrix} = (\pm 1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{So } x = (\pm 1) [d(ax+by) - b(cx+dy)]$$

$$y = (\pm 1) [-c(ax+by) + a(cx+dy)]$$

This implies that x and y are both linear combinations of $(ax+by)$ and $(cx+dy)$

So any common divisor of $(ax+by)$ and $(cx+dy)$ is also common divisor of x and y

So $\gcd(ax+by, cx+dy)$ is a common divisor of x, y . ①

And since $ax+by, cx+dy$ are linear combinations of x, y . By the same logic, $\gcd(x, y)$ is a common divisor of $ax+by$ and $cx+dy$. ②

By ①, ②, $\gcd(x, y) = \gcd(ax+by, cx+dy)$. 🍌

Questions assigned to the following page: [3](#) and [4.1](#)

$\overline{T}$
 To show this, denote $ax+by$ as m and $cx+dy$ as n . We know $\gcd(m,n) \mid x$, $\gcd(m,n) \mid y$, so $\gcd(m,n) \leq \gcd(x,y)$
 Same we have $\gcd(x,y) \leq \gcd(m,n)$
 So $\gcd(m,n) = \gcd(x,y)$.
 Then we have finished the proof.

4. For any integer m , we can use the Fundamental Theorem of Arithmetic to write $m = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ where the p_i 's are distinct primes in an (essentially) unique way. The natural number a_i is said to be the multiplicity of the prime p_i in m . [By convention, the multiplicity of p in m is 0 if p does not divide m .]

1. Let d and n be positive integers. Prove that n is a d -th power of some other integer if and only if for every prime p , the multiplicity of p in n is divisible by d . ①
2. Prove that if n is not a d -th power of some other integer, then $\sqrt[d]{n}$ is irrational. [Hint: try proof by contradiction.] ②

1. Proof Note the two statements as ①, ② respectively.
 First, we prove $\textcircled{1} \rightarrow \textcircled{2}$

(1.) Let $n = \beta^d$ for some integer β .
 Let p be an arbitrary prime number.

We want to show that

for some integer k , $p^k \mid n$ and $d \mid k$.

Question assigned to the following page: [4.1](#)

By FTA we know $\beta = q_1^{a_1} q_2^{a_2} \dots q_m^{a_m}$ for
some primes $q_1, \dots, q_m$ and integers
 $a_1, a_2, \dots, a_m$

$$\text{Then } n = q_1^{a_1} q_2^{a_2} \dots q_m^{a_m}$$

We prove the statement by cases.

Case 1 $p = q_i$ for some $1 \leq i \leq m$.

Then consider $k = a_i d$

$$\text{We know } n = q_i^{a_i} \cdot (q_1^{a_1} \dots q_{i-1}^{a_{i-1}} \cdot q_{i+1}^{a_{i+1}} \dots q_m^{a_m})$$

$$\text{So } q_i^{a_i d} \mid n, \text{ that is, } \underline{p^k \mid n}$$

Also, since $k = a_i d$, $\underline{d \mid k}$

This completes the proof of $\textcircled{1} \rightarrow \textcircled{2}$

Next we prove $\textcircled{2} \rightarrow \textcircled{1}$

(2) Let p be an arbitrary prime, n be
an arbitrary integer with the property
that for any prime p , the multiplicity of
 p in n is some integer a s.t. $a = kd$ for
some integer k .

Questions assigned to the following page: [4.2](#) and [4.1](#)

By FTA, we know $n = p_1^{a_1} p_2^{a_2} \dots p_m^{a_m}$

for some primes $p_1, p_2, \dots, p_m$ and their multiplicity $a_1, a_2, \dots, a_m$

And by the property above we have

$a_1 = k_1 d, a_2 = k_2 d, \dots, a_m = k_m d$ for some integers $k_1, k_2, \dots, k_m$

Then $n = (p_1^{k_1} p_2^{k_2} \dots p_m^{k_m})^d$.

Thus there is always an integer

$(p_1^{k_1} p_2^{k_2} \dots p_m^{k_m})$ such that n is its d -th power.

This completes the proof of $(2) \rightarrow (1)$

Since we have $(1) \rightarrow (2)$, $(2) \rightarrow (1)$, we have proved the if-and-only-if relation between the two statements.

2 - Proof We prove it by contradiction.

Let n be an arbitrary integer s.t. n is not a d -th power of any integer,

Assume for contradiction that $\sqrt[d]{n}$ is a rational number. $d \in \mathbb{Z}^+$.
So $\sqrt[d]{n} = \frac{p}{q}$ for some
 $p, q \in \mathbb{Z}$ with $q \neq 0$, where $(p, q) = 1$.

Question assigned to the following page: [4.2](#)

Then $n = \frac{p}{q^d}$, $nq^d = p$.

By FTA, $q = q_1 q_2 \dots q_m$ for some primes 1

$$p = p_1 p_2 \dots p_s \text{ for some primes } q_1, \dots, q_m$$

Then $n (q_1^d q_2^d \dots q_m^d) = p_1^d p_2^d \dots p_s^d$ $p_1, \dots, p_s$

Since $(p, q) = 1$, none of $p_1, \dots, p_s$ is equal to any of $q_1, \dots, q_m$. 2

So n must contain the factor $p_1^d p_2^d \dots p_s^d$.

And since $n (q_1^d q_2^d \dots q_m^d) = p_1^d p_2^d \dots p_s^d$,
where $p_1, \dots, p_s$ are all primes, any prime factor of n must be among $p_1, p_2, \dots, p_s$.

So all prime factors of n have the multiplicity of d , then by part I we know that n is the d^{th} power of some integer, which contradicts the condition.

Then we have completed the proof.

No questions assigned to the following page.

5. **We will see many equivalence relations in this course, and we will apply this problem many times during the semester.** Let X be a non-empty set, and let $\sim$ be an equivalence relation on the set X , that is, a rule used to compare pairs of elements of X which satisfies the following properties (we read “ $a \sim b$ ” as “ a is related to b ”):

- $\sim$ is reflexive: $a \sim a$ for all $a \in X$.
- $\sim$ is symmetric: let $a, b \in X$ such that $a \sim b$. Then $b \sim a$.
- $\sim$ is transitive: let $a, b, c \in X$ such that $a \sim b$ and $b \sim c$. Then, $a \sim c$.

For example, before you start this problem note that the relation $=$ (being equal to) is an equivalence relation on any non-empty set X . However, the relation $\leq$ (being less or equal to) on the set $\mathbb{R}$ is not an equivalence relation, as it is not symmetric. Determine whether the following rules are equivalence relations on the given sets.

(a) Let X be $\mathbb{R}^2$, and $\sim$ is the rule

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} w \\ z \end{bmatrix}$$

if $x - y = w - z$.

(b) Let X be the set of integers, $\mathbb{Z}$, and let $\sim$ be the rule

$$a \sim b$$

if $ab \geq 0$.

(c) Given an equivalence relation $\sim$ in X , we define the **equivalence class of a** (denoted by $[a]$) for all $a \in X$ as

$$[a] := \{x \in X : x \sim a\} \subseteq X.$$

Show that, for all $a \in X$, $[a] \neq \emptyset$.

(d) Let $a, b \in X$. Show that either $[a] \cap [b] = \emptyset$ or $[a] = [b]$.² That is, prove that **equivalence classes are either disjoint or equal**.

(e) Let $\mathcal{S} := \{[a] : a \in X\}$ the set of all equivalence classes with respect to the relation $\sim$. Prove that

$$X = \bigsqcup_{Y \in \mathcal{S}} Y,$$

(the symbol $\sqcup$ means *disjoint union*). In other words, prove that **equivalence classes partition X into disjoint sets**.

²Hint: Suppose that $[a] \cap [b] \neq \emptyset$, that is, there exists $c \in X$ such that $c \in [a] \cap [b]$. Do you remember how to prove “or” statements?

Questions assigned to the following page: [5.1](#) and [5.2](#)

(a) Let $\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} w \\ z \end{bmatrix} \in \mathbb{R}^2$ be arbitrary, with $x-y = w-z$

Since $x-y = x-y$,

$\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} x \\ y \end{bmatrix}$, meetingy reflexive.

Since $x-y = w-z$, $w-z = x-y$

This implies $\begin{bmatrix} w \\ z \end{bmatrix} \sim \begin{bmatrix} x \\ y \end{bmatrix}$ whenever $\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} w \\ z \end{bmatrix}$,
meetingy symmetric

Let $\alpha, \beta \in \mathbb{R}$ be arbitrary with $w-z = \alpha - \beta$

Since $x-y = w-z$, $\begin{bmatrix} w \\ z \end{bmatrix} \sim \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

We have $x-y = \alpha - \beta \Leftrightarrow \begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

This implies transitive

So the rule is a equivalence relation

(b) For transitivity, consider $a=1$, $c=-1$, $b=0$

Then $ab=0$, so $a \sim b$

$bc=0$, so $b \sim c$

But $ac=-1 < 0$, $a \not\sim c$

In this case $a \sim b$, $b \sim c$ and $a \not\sim c$

So the rule does not meet transitive.

Therefore it is not an equivalence relation.

Questions assigned to the following page: [5.4](#) and [5.3](#)

cc) Let a be an arbitrary element of X .

By reflexive property of equivalence relation,

$a \sim a$ and since we know $a \in X$,
 a must be in $\{x \in X \mid x \sim a\}$

So the set contains at least one element,

Therefore for all $a \in X$, $[a] \neq \emptyset$ since our selection of a is arbitrary.

(d) Assume $[a] \cap [b] \neq \emptyset$

Then $\exists x \in X$ s.t. $x \in [a]$ AND $x \in [b]$

Let m be an arbitrary element of $[a]$, by definition we have $m \sim a$

since $x \sim a$ AND $m \sim a$, by transitive we have
 $m \sim x$.
(can't be symmetric)

Since the selection of m is arbitrary,

$$[a] = \{x \in X \mid x \sim a\}$$

Questions assigned to the following page: [5.4](#) and [5.5](#)

Similarly, let n be an arbitrary element of $[b]$,

We will see $n \sim x$, so $[b] = \{x \in X \mid x \sim n\}$

So $[a] = [b]$

Therefore either $[a] \cap [b] = \emptyset$ or $[a] = [b]$.

(e) Let x be an arbitrary element of X .

By reflexive property of equivalence class,

$x \sim x$, and thus $[x]$ is non-empty since it at least contains x

So $[x] \in S$.

Therefore $\forall x \in X, [x] \in S$

Hence $X = \bigcup_{Y \in S} Y$

And by (d) we have proved that $\forall x, y \in X$

either $[x] = [y]$ or $[x] \cap [y] = \emptyset$

So by definition, $\forall [x], [y] \in S$, they are disjoint.

So $X = \bigsqcup_{Y \in S} Y$ by definition.